

Finding new modulators for the THIK-1 channel using computational methods

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References:

K. E. Rödström, et al. *Nature Structural & Molecular Biology*, 2025.
R. W. Bürli, et al. *ACS Medicinal Chemistry Letters*, 2024, **15**, 646–652.

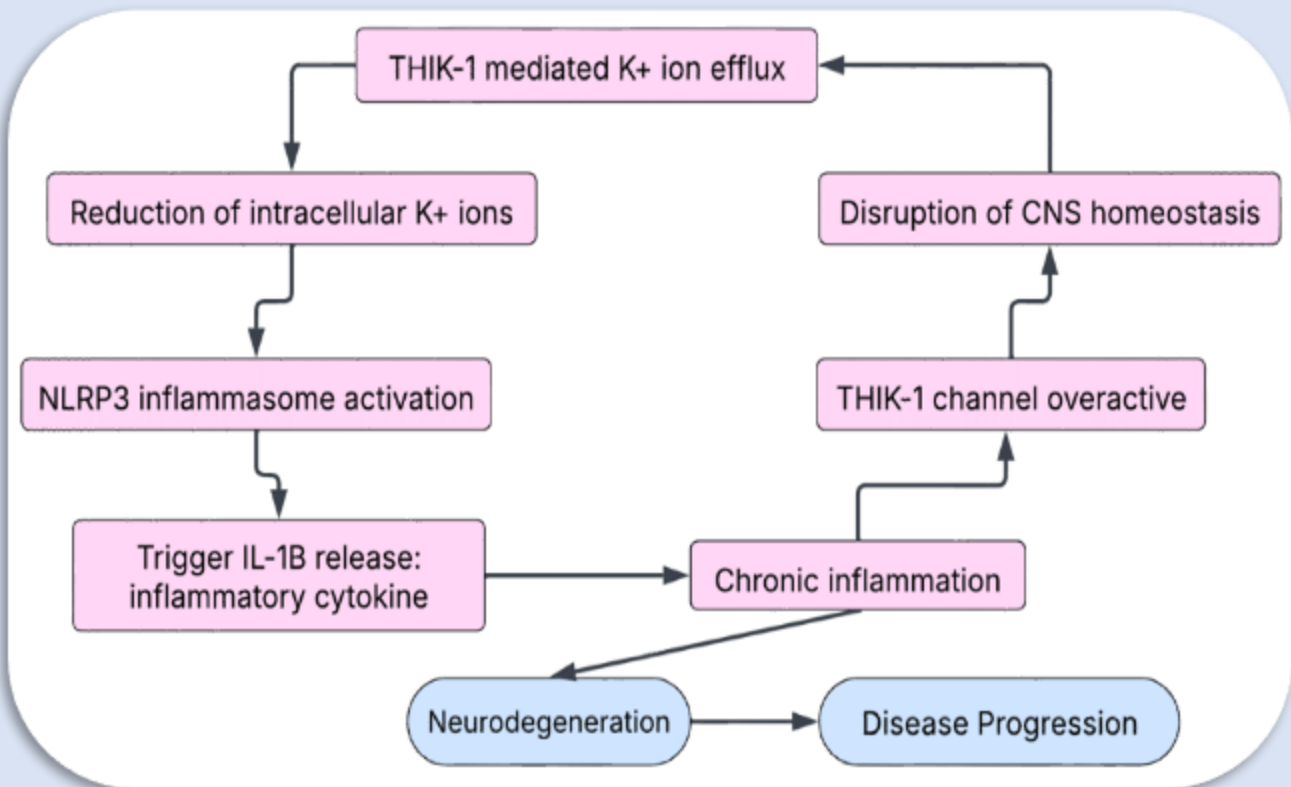
INTRODUCTION

THIK-1, a two-pore domain potassium ion (K⁺) channel impacts **immune response** in the brain, offering a potential therapeutic target for neurodegenerative disorders such as **Alzheimer's and Parkinson's disease**.
The **cryo-EM structure** of the protein has **only recently been unveiled**. Molecular docking of a known **THIK-1 modulator CVN293** investigates binding mechanisms with the aim to **understand how THIK-1 can be targeted by small molecule inhibitors for therapy**.

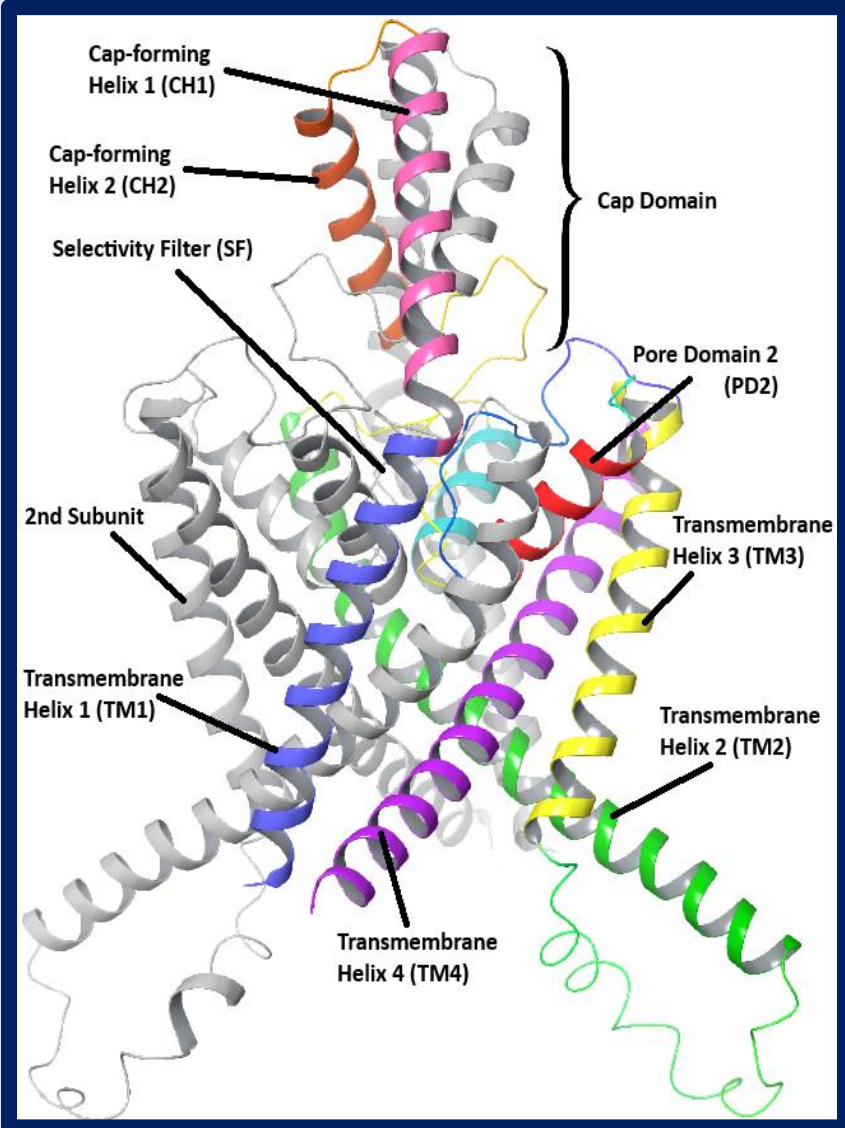
THIK-1 SIGNIFICANCE

- THIK-1 is expressed in **microglia**, the **CNS immune cells**.
- Microglia** maintain brain health and **respond to damage**.
- The **IL-1B** gene **accelerates neurodegenerative disease**.

NEURODEGENERATION THIK-1 CYCLE

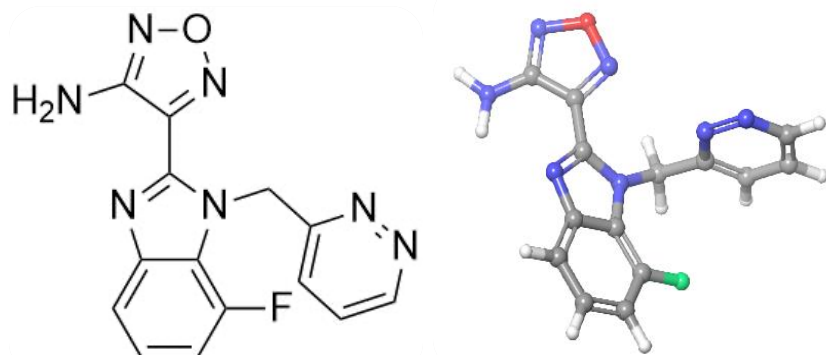


THIK-1



CVN293: A THIK-1 MODULATOR

CVN293 is a reported small-molecule THIK-1 inhibitor, discovered before the crystal structure of THIK-1 was unveiled. Its **binding mechanism is unknown**.



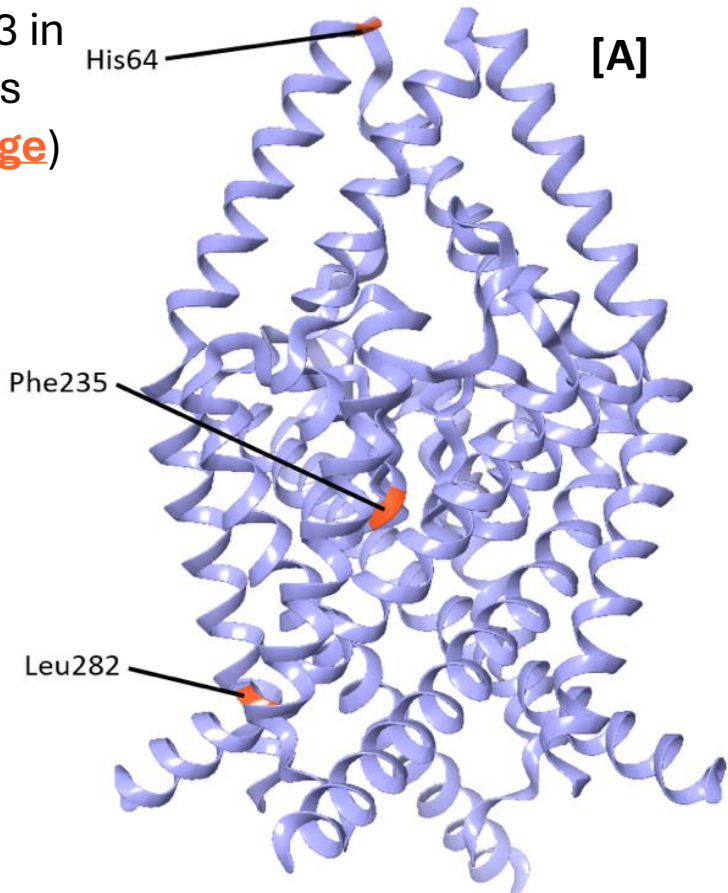
INITIAL INVESTIGATION

We **docked** CVN293 in different grid regions (highlighted in **orange**) of THIK-1 that coordinated to **key features** of the channel. **[A]**

His64: Cap domain

Phe235: SF

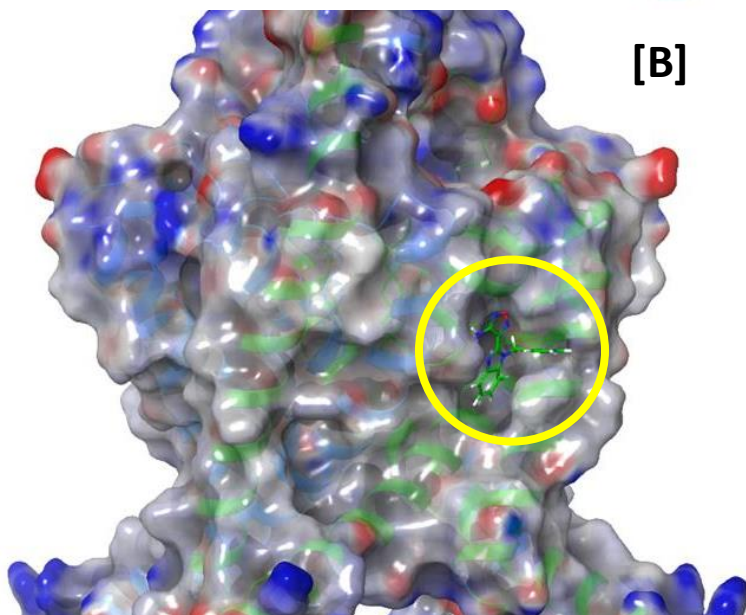
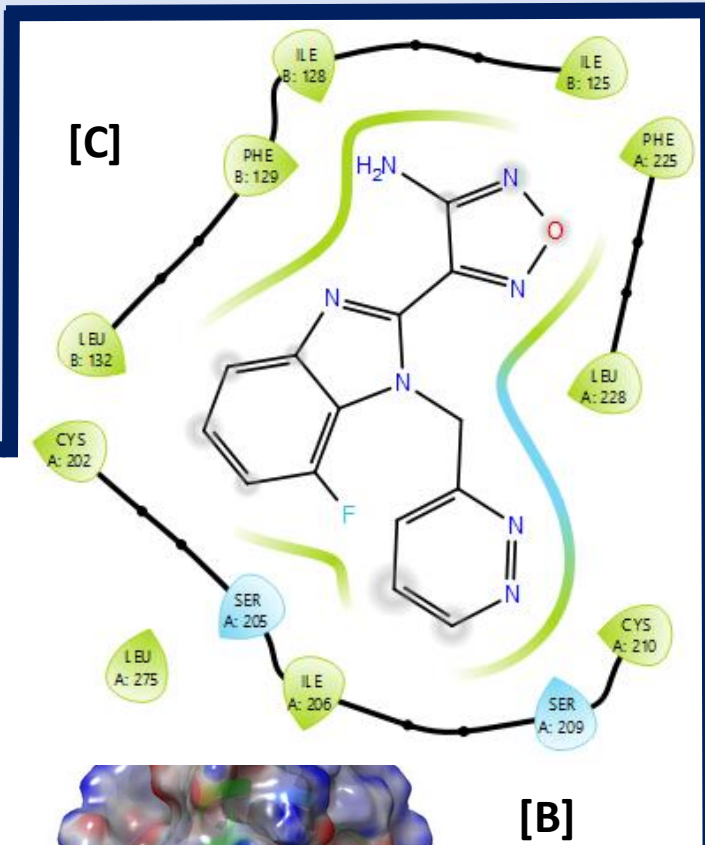
Leu282: TM4



Docking produced '**poses**' within the protein and analysis of **interaction diagrams** highlighted key interactions.

The strongest pose, docked into THIK-1 **[B]**, produced a **docking score of -5.453** however showed only **non-specific hydrophobic interactions, [C]**.

Poses	Residues + Docking Scores		
	64 His	235 Phe	235 Leu
Pose 1	-4.629	-5.543	-4.539
Pose 2	-4.183	-4.997	-4.379
Pose 3	-3.599	-4.874	-3.945
Pose 4	-3.477	-3.993	-3.646
Pose 5	-3.115	-	-3.422

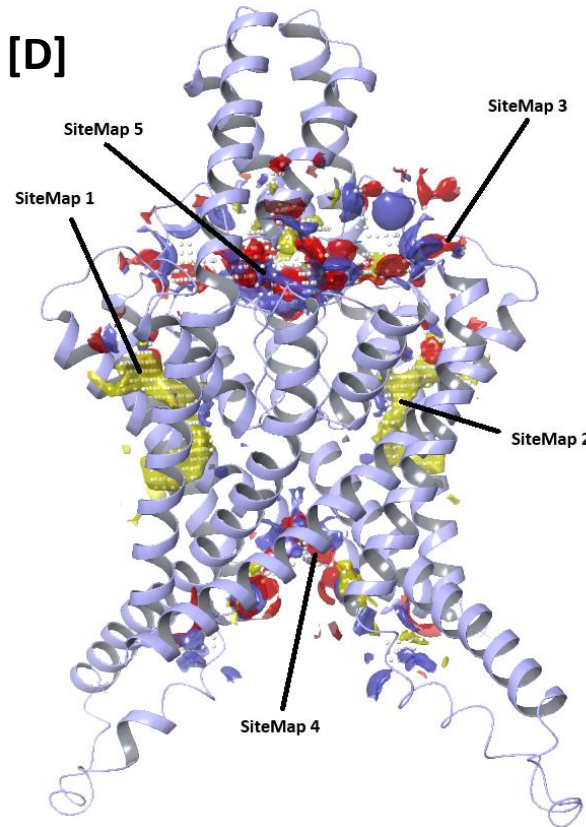


SITEMAP ANALYSIS

SiteMap analysis produced **5 potential binding regions** in THIK-1, **[D]**. Site 1 and 2 correspond to a known lipid binding site.

Docking of CVN293 in the 5 highlighted regions produced the second strongest pose with a **docking score of -5.141**, located in **Site 1, [E]**.

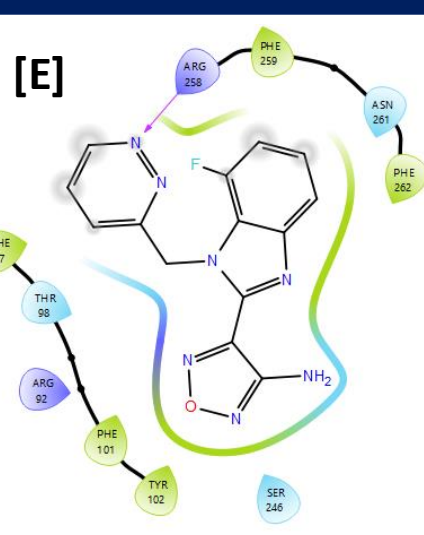
SiteMap	SiteScore	Size	Pose 1
1	1.244	198	-5.141
2	1.230	205	-4.246
3	1.044	348	-2.080
4	1.037	110	-4.149
5	0.949	75	-2.555



CONCLUSIONS

A total of **37 poses** were generated. We are reporting this as the first docking of THIK-1 to have been conducted.

The **Table** below indicates that **CVN293 shows no typical binding mechanism**. Our findings cannot conclude whether a single binding site exists between CVN293 and THIK-1.



Type of Interaction	No. of Poses
Hydrogen Bonding	10
Pi-Cation	6
Pi-Pi Stacking	9
No Key Interaction	12

FUTURE RESEARCH & APPLICATIONS

To further investigate the binding mechanism of CVN293, modulation of THIK-1 by **site-directed mutations** at key residues within the identified binding regions could hold valuable in **determining specific residue impact on binding affinity**.

Docking structural analogues of CVN293 within the same parameters can aid in therapeutic small molecule optimisation of the channel.

The **continued investigation** on the lesser known K2P channel is key to understanding **THIK-1's significance in therapeutic modulation**.